

RAPPORT DE STAGE 5BIM

Relationship between environment and sole recruitment in the Bay of Biscay

IAN PELLET

Département Biosciences, BIM (Bioinformatique et modélisation)

INSA (Institut national des sciences appliquées) de Lyon

Maître de stage

JEAN BAPTISTE LECOMTE

DECOD (Dynamique et durabilité des écosystèmes),

Institut Agro - Ifremer - Inrae

OLIVIER LE PAPE

DECOD (Dynamique et durabilité des écosystèmes),

Institut Agro - Ifremer - Inrae

Enseignant référent

SANDRINE CHARLES

LBBE (Laboratoire de Biométrie et Biologie Evolutive),

Université Claude Bernard Lyon 1

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1 Introduction

The common sole (*Solea solea*) of the Bay of Biscay (ICES Area VIIIa/b) is a nursery-dependant species of high commercial value for the French fisheries of the Atlantic coast. This stock is assessed by the International Council for the Exploration of the Sea (ICES). In 2022, the total admissible catch (TAC) for this stock has been reduced by 36% compared to 2021 (ICES 2023b). This important reduction of the TAC is mainly due to the decline of the recruitment of young soles in the fishing stock estimated by the stock assessment model since 2009. This decline in recruitment implies a decrease in the stock biomass and catches, independently from the fishing pressure. The stock is subjected to a restrictive regulatory framework but the decrease in fishing mortality is not sufficient to improve the loss of biomass and insure a sustainable exploitation of the stock (ICES 2023a).

The sole in the Bay of Biscay spawns in late winter and spring, with a peak of the spawning period in March (Rijnsdorp et al. 1992; Guillou and Njock 1978; Arbault, Camus, and Bec 1986). In the Bay of Biscay, sea water temperature does not seem to affect the timing of spawning (Koutsikopoulos and Lacroix 1992). The main spawning activity occurs offshore at depths from 30 to 100 m (Koutsikopoulos, Karakiri, et al. 1989; Arbault, Camus, and Bec 1986). Larvae hatch after about 7 to 8 days and settle at the bottom during metamorphosis about 3 weeks after hatching (Riley 1974; Fonds 1979). The duration of the planktonic phase is normally about 4 weeks. During the planktonic phase, the eggs and larvae are exposed to the residual currents. As the duration of the planktonic phase is relatively short, the surviving post-larval soles usually settle close to the spawning area. Nursery areas are found in shallow sandy or muddy coasts with a substrate characterised by a particle size of 4 to 500 μm and a depth down to 20 m, although they are most abundant at depths of less than 10 m (Koutsikopoulos, Karakiri, et al. 1989; Marchand 1991; Rijnsdorp et al. 1992).

Nursery-dependent populations, such as the sole, are subjected to different pressures during their life cycle. These pressures can be of an environmental nature (variation in climate) or of an anthropic nature (fishing, habitat quality). The different life stages are not subjected to the same pressures: while the adult individuals are mainly impacted by fishing activities, the survival during the first pelagic state (eggs and larvae) is mainly determined by the hydroclimatic conditions (Savina et al. 2016). The extent and quality of coastal nurseries affect the juvenile stages and determine the recruitment of the

population, and thus the state of the stock.

For decades, coastal nurseries have been affected by human activities and global changes. Soles are the most susceptible to changes in the environment during their first year and this susceptibility decreases with age (Koutsikopoulos, Karakiri, et al. 1989). During this period, changes in the quality and surface of nursery habitats can have an important impact on the future recruitment of soles (Rijnsdorp et al. 1992; Le Pape, Chauvet, et al. 2003; Gibson 1994; Champagnat et al. 2021; Rochette et al. 2010; Archambault et al. 2018). Those phenomena have been studied for the sole in the Eastern English Channel (Archambault et al. 2018; Champagnat et al. 2021; Rochette et al. 2010), where successive research projects allowed for a better understanding of the effects of changes in the extent and quality of coastal nurseries at the stock scale. Sole nurseries in the Bay of Biscay have been identified (Le Pape, Chauvet, et al. 2003; Trimoreau et al. 2013; Regimbart, Guitton, and Le Pape 2018) and the important role of river flow and nutrient inputs from the rivers on the development of juveniles in these nurseries, and thus on the variation in recruitment, have been highlighted (Le Pape, Désaunay, and Guérault 2003; Kostecki et al. 2010; Pape et al. 2013).

The first aim of this study is to characterise the effect of river flow on age 0 soles in the nurseries of the Bay of Biscay. To do so, a long time series of data from a scientific survey carried out on the nursery grounds of the Bay of Biscay is used to analyse the effect of river flow as well as the effect of the habitat type on the abundance of soles juveniles. The results of this first analysis are used to integrate the effect of river flow on recruitment into the management of the stock. Several management scenarios are defined using different strategies to integrate the river flow effect into the stock assessment model. In order to test and compare these scenarios to the current assessment method used by ICES, a management strategy evaluation (MSE) framework is developed.

2 Materials and methods

2.1 Modelling the effect of river flow on juvenile abundance

2.1.1 Survey data

NURSE is a set of scientific fishing surveys performed in the Bay of Biscay's coastal and estuarine nursery grounds of benthic-demersal fishes (Brind'Amour and Desaunay 1981). The sampling is done in waters of a maximal depth of 40m in six zones: Loire and Gironde estuarine areas, Vilaine Bay, Bourgneuf Bay, Pertuis d'Antioche and Pertuis Breton. The hauls are performed using a beam trawl. The surveys started in 1980 and have been ongoing since. The Vilaine Bay has been sampled almost every survey year since 1982, and the other zones are sampled in alternation each year.

The NURSE surveys follow a unified protocol to ensure sampling homogeneity (De-launay and Brind'Amour 2018). Some hauls are removed from the analysis because, in the first surveyed years, a few hauls have been performed using trawls with tickler chains. The tickler chains increase the catch rate, so the data from these hauls cannot be used without adjustment and is removed from the data set. Another potential source of variability caused by sampling variables is the period of the year during which the hauls are performed. The young soles migrate out of the estuaries in autumn (Dorel et al. 1991), so the hauls performed after the 15th of October are removed. Table 1 summarises the number of hauls in each zone and each year once the data have been filtered.

The first year of the sole life being the one during which they are the most susceptible to environmental changes (Koutsikopoulos, Karakiri, et al. 1989), the analysis are run only for 0 age group (G0). The NURSE survey targets small fishes and a high proportion of G0 soles is present in the data set. The age of soles caught is inferred according to the maximal length per age group per month established by otolith reading (personal communication Brind'Amour). The data used in the following analysis is the number of G0 sole per hauls.

In order to account for variability in fishing effort, the number of soles caught by age and haul is normalised by the swept area of each haul. This normalised data is the catch per unit of effort (CPUE) defined in the equation (1).

$$CPUE = \frac{\text{number of individuals}}{\text{swept area}} \quad (1)$$

	Bourgneuf	Gironde	Loire	Pertuis Antioche	Pertuis Breton	Vilaine	Total
1981	5	-	-	-	-	-	5
1982	9	-	8	-	-	-	17
1983	-	-	10	-	-	-	10
1984	-	-	23	-	-	-	23
1985	-	-	-	-	-	46	46
1986	-	-	22	21	52	47	142
1987	-	-	-	16	19	72	107
1988	-	-	-	-	-	57	57
1989	-	-	-	-	-	57	57
1990	-	-	-	-	-	45	45
1991	-	-	-	-	-	-	-
1992	-	-	-	-	-	48	48
1993	-	-	-	-	-	48	48
1994	-	-	-	-	-	-	-
1995	-	-	-	-	-	-	-
1996	-	22	-	42	29	30	123
1997	14	38	21	-	2	48	123
1998	-	-	-	-	-	-	-
1999	-	-	-	-	-	-	-
2000	19	23	26	19	21	31	139
2001	15	36	23	25	24	57	180
2002	11	18	20	13	13	20	95
2003	11	22	19	15	11	22	100
2004	-	-	-	-	-	42	42
2005	-	-	-	-	-	42	42
2006	-	-	-	-	-	-	-
2007	-	-	-	-	-	-	-
2008	24	-	24	-	-	52	100
2009	-	30	-	-	-	50	80
2010	-	-	13	-	-	39	52
2011	-	-	-	-	-	-	-
2012	-	-	21	-	-	51	72
2013	-	-	-	-	-	-	-
2014	21	-	-	-	-	50	71
2015	-	-	-	30	30	50	110
2016	30	26	46	-	-	-	102
2017	-	-	-	30	30	50	110
2018	11	27	28	-	-	50	116
2019	-	35	-	50	43	51	179
2020	11	25	20	29	28	44	157
2021	-	10	-	45	42	36	133
Total	181	312	324	335	344	1235	2731

Table 1: Number of hauls per year for each sampled zone, after filtering hauls performed after the 15th of October and hauls performed with a tickler chain.

2.1.2 River flow data

To study the effect of river flow on G0 soles in nurseries, the flow data from rivers inflowing in the samples nursery areas is used. The rivers considered are the Vilaine River for the Vilaine Bay, the Loire River for the Loire estuary and the Bourgneuf Bay, the summed flow of Garonne and Dordogne for the Gironde estuary, the Charente for the Pertuis Antioche and the Sèvre Niortaise for the Pertuis Breton. The mean monthly flow of these six rivers is retrieved from Banque Hydro, the national hydrology data bank of the French Ministry of the Environment (Anonymous n.d.). The data bank provides data from several hydrological stations on each river. The selected hydrological station for each

studied river is the station the most downstream possible with the longest time series of data matching the NURSE survey data (appendix A).

A preliminary analysis of seasonal variations of each of the rivers was done to select the period in which the variability of flow is most likely to impact the juvenile sole abundance. The period from January to May is kept as it is a period with higher flow and interannual variability as well as the period right before the settlement of young soles in nurseries (Le Pape, Désaunay, and Guérault 2003).

The mean flow of the different rivers considered are not of the same magnitude (appendix B). In order to normalise the flow data across the different rivers the flow of each river is summarised by year as an anomaly of flow during the period January-May (figure 1). The mean flow of each river is computed for each year and noted $Qm_{y,r}$ with y the year and r the river. This yearly measure of flow is normalised by the mean of the flow values of the river from 1980 until 2022, $\mu_r^{1980-2022}$ and the standard deviation $\sigma_r^{1980-2022}$ to get the river anomaly for the year:

$$Qa_{y,r} = \frac{Qm_{y,r} - \mu_r^{1980-2022}}{\sigma_r^{1980-2022}} \quad (2)$$

2.1.3 Bathymetry data

The main data set used to assign a bathymetry value to each haul is the EMODnet Bathymetry Digital Terrain Model (Anonymous 2022a). However, some of the hauls locations are not available in the EMODnet data set, especially in the Loire estuary, upstream of the Saint-Nazaire bridge. The bathymetry data set from GEBCO: GEBCO_2022 Grid (ice surface elevation) (Anonymous 2022b) is used to complete the missing data. Despite some missing areas, the EMODnet data set is chosen as the main data source because of its finer spatial resolution of 1/16 arc minute against 1/4 arc minute for the GEBCO grid.

2.1.4 Sediment data

The sediment type at the hauls location is taken from the Sextant Ifremer mapping of physical habitat (Hamdi, Vasquez, and Populus 2010). The sediments are classified according to EUNIS (European Nature Information System) typology. EUNIS categories are grouped as described in table 2 to produce homogeneous habitat categories for flatfish

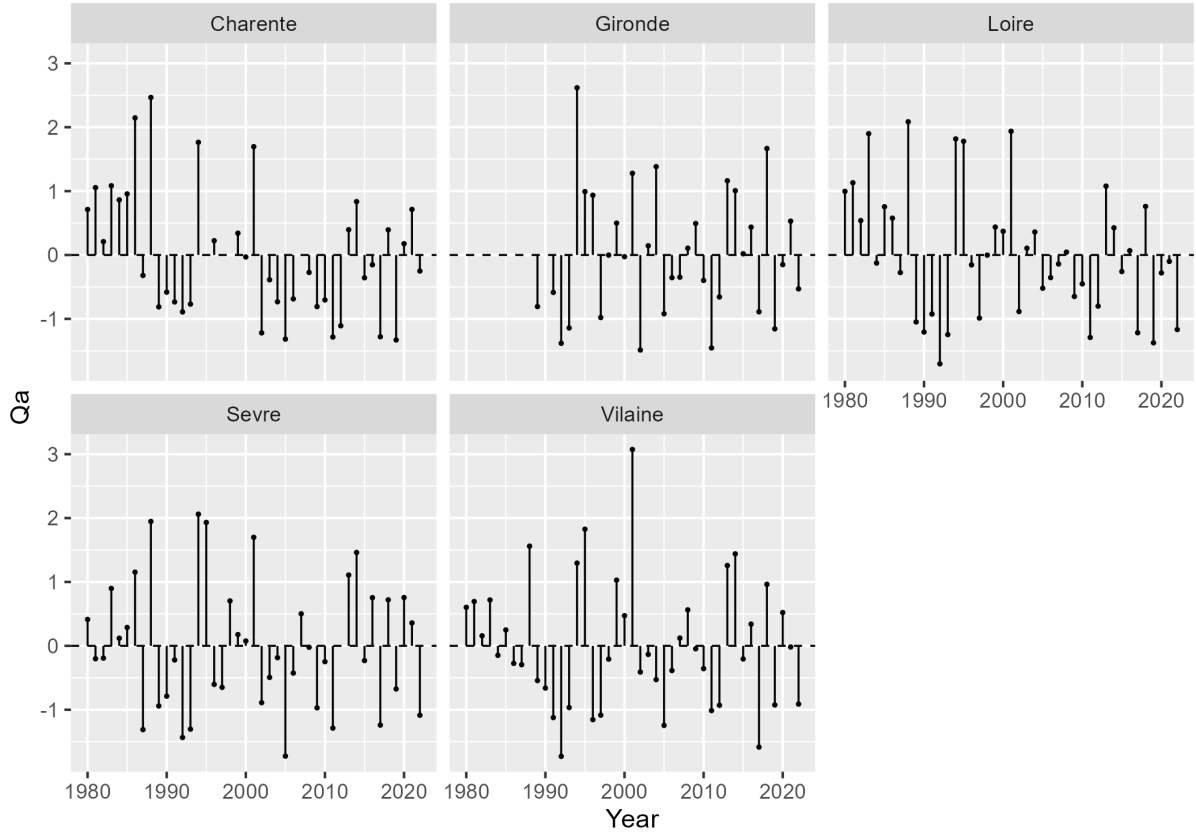


Figure 1: River flow anomaly Q_a for each river over the period 1980-2022.

Sediment type	Eunis ID
coarse sediment	A5.14, A5.13, A5.15, A3.2, A1, A4.3, A3.3, A3.1 , A4.1, A4.2
mixed sediments	A5.44, A5.45, A5.43
mud	A5.34, A5.36, A2.3, A5.33, A5.35, A5.37
sand	A5.23, A5.25, A5.26, A5.24, A5.27, A2

Table 2: New sediment categories created from Eunis categories.

species. Some of the survey hauls were performed at a location where no sediment type is available in this database. These missing data points are mainly present upper in the estuaries. Sediment types are assigned to these points according to the closest area with known sediment type.

2.1.5 Delta model

The catch data transformed to CPUE contains a high percentage of zeros because of the high number of hauls without any young sole caught, but it can also have very high values of CPUE (figure 2). The presence of many zeros can lead to issues in computation and is

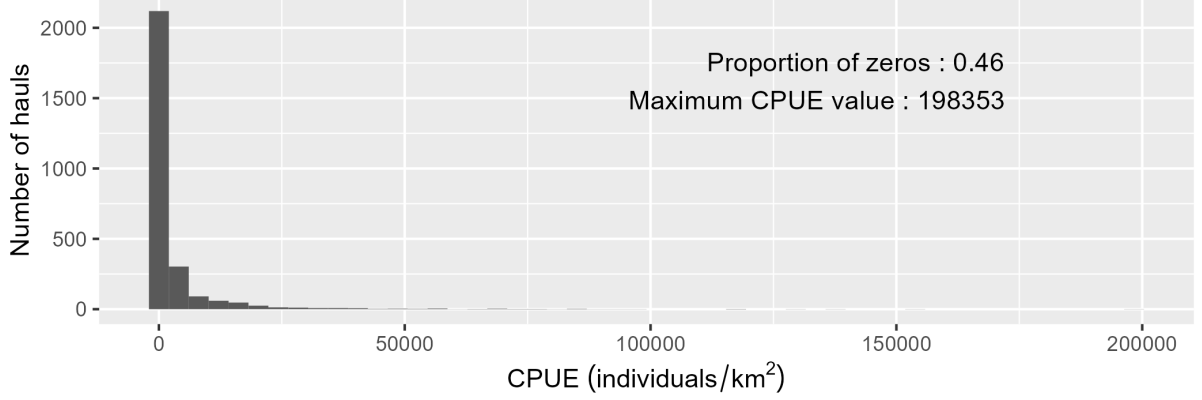


Figure 2: CPUE values distribution for age 0 group of sole in the filtered data-set.

not adequately modelled by classical general linear model distributions. The delta model solves the issues encountered with classical linear models when modelling data with a high percentage of zeros by using two sub-models, one binomial model describing the presence/absence phenomenon and a second one modelling only the positive catch data. The two sub-models can reflect two different processes with different explanatory variables and the positive density model can follow any strictly positive distribution. In our case, the positive CPUE values are log-transformed and modelled by a normal distribution :

$$\Delta : \begin{cases} presence \sim Binomial(p), & logit(p) = \alpha X + \epsilon_1 \\ log(CPUE_{>0}) \sim N(\mu, \sigma), & \mu = \beta Y + \epsilon_2 \end{cases} \quad (3)$$

with α and β the vector of fitted coefficients, X and Y the explanatory variables and ϵ_1 and ϵ_2 the residual error of the model. The fitted delta-lognormal model can be used for prediction by recombining the probability of presence and the mean strictly positive CPUE with a correction term to obtain an unbiased estimate from a linear model based on log-transformation (Laurent 1963) :

$$predict_{\Delta}(X, Y) = p \times e^{\mu} \times e^{\frac{\sigma^2}{2}} \quad (4)$$

The effects of the variables provided in table 3 are included as explanatory variables to select the most appropriate model to describe the abundance of young sole in the nurseries of the Bay of Biscay. First, models with the different combinations of the variables apart from the flow anomaly Qa are fitted and their relative quality is compared using the Akaike information criterion (AIC). The model with the lowest AIC is used as a base model to test the effect of river flow anomaly and its interaction with the nursery zone.

Variable	Description	Type	Modalities
sediment	type of sediment at haul location	categorical	coarse sediment, mixed sediment, mud, sand
bathymetry	bathymetry at haul location	categorical	<-20m, [-20,-15], [-15,-10], [-10,-5], >-5m
zone	zone of the haul	categorical	Vilaine, Loire, Gironde, Bourgneuf, Pertuis Antioche, Pertuis Breton
Qa	flow anomaly of the hatching year	continuous	

Table 3: Description of the variables tested to explain the variability of observed G0 soles abundance.

To determine whether or not there is a statistically significant difference between the base model and the models with added terms Qa and $zone \times Qa$, a χ^2 test for the binomial distribution or F test for the Gaussian distribution is used, with a type I error of 5%.

2.2 Management Strategy Evaluation (MSE)

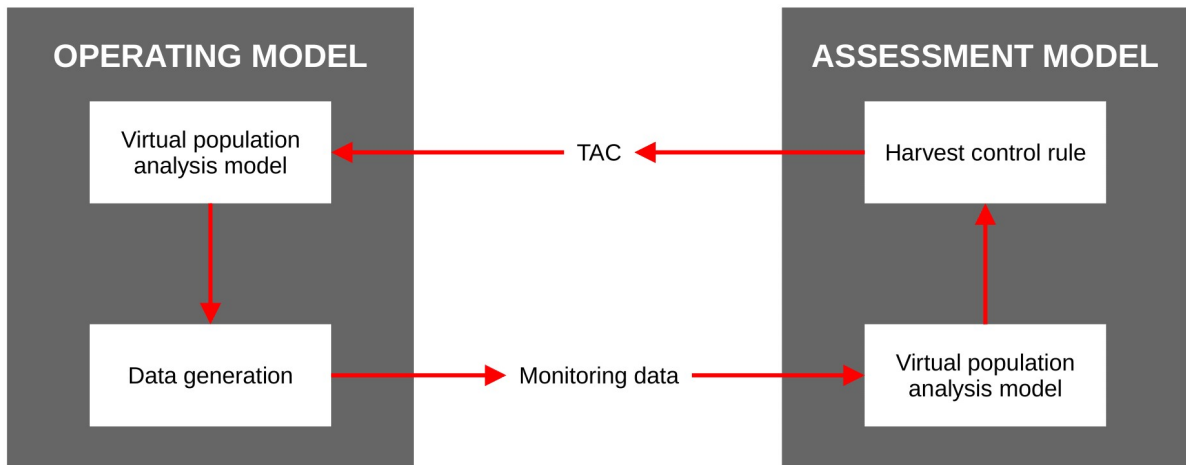


Figure 3: Conceptual overview of the management strategy evaluation process. The operating model part is composed of a virtual population analysis (VPA) model that simulates the stock and a data generation process that provides monitoring data (indices, catches...) to the assessment model. The assessment model is also based on a VPA model to estimate the state of the simulated stock and uses a harvest control rule to set the advice as total admissible capture (TAC) for the next year. This TAC value is used by the operating model to simulate the catches.

A Management Strategy Evaluation (MSE) framework is developed to compare different harvest control rules (Punt et al. 2016). The MSE consists of simulating a fish stock, applying different harvest control rules and evaluating the consequences on performance

metrics such as spawning stock biomass (SSB) or catches. It comprises two models: (i) the operating model, which is a mathematical representation of the biological-fishery system and (ii) a stock assessment model with a harvest control rule, which represents the management regulation applied to the system (figure 3). The two models inter-operate through a feedback loop. The operating model provides monitoring data that is used by the assessment model and the assessment model sets the management regulations that will be applied to the fishing activity that is implemented in the operating model.

2.2.1 Assessment model

The assessment model is a virtual population analysis (VPA) as the one used by ICES to perform the stock assessment (ICES 2023b; Shepherd 1999). It takes as input the commercial catches, ages and length frequencies from catch sampling, a fixed maturity ogive estimated in 2000, natural mortality assumed to be fixed at 0.1, one scientific survey index, the French Bay of Biscay sole beam trawl survey (ORHAGO), and four commercial indices (FR-SABLES, FR-ROCHELLE, FR-BB-IN-Q4 and FR-BB-OFF-Q2) (ICES 2023b; Majewska 2023). The commercial catches and the indices are derived from the simulation of the operating model, the other data stays constant over time and is the data used by ICES (ICES 2023b).

Generating index time series

In order to run the assessment, index time series have to be derived from the stock number at age of the operating model for each year of the simulation. To do so, each index time series is fitted to the number at age estimated by the VPA using a linear model (equation 5) (ICES 2023b).

$$idx_{y,a} \sim age + stock_{y,a} + age \times stock_{y,a} \quad (5)$$

The fitted model is used to derive an index for each age class according to the number at age of the stock for each projected year. Uncertainty is introduced in the index time series by drawing indices values in a normal distribution centred on the mean index and standard error predicted by the linear model.

Stock projection

For each year y , the assessment model is fitted to data until year $y - 1$. The advice is then

given for year $y + 1$ as a total allowable catch (TAC). To give the advice, the assessment model is used to project the stock for years y , $y + 1$ and $y + 2$ with assumptions on recruitment for year $y + 1$ and $y + 2$ and assumptions on fishing mortality (F) in years y and $y + 1$. The assumption on recruitment is that it is equal to the geometric mean of the three previous years. The F of year y is set as the F of year $y - 1$. The F for the year $y + 1$ is chosen following a harvest control rule.

2.2.2 Operating model

The operating model is a VPA model that uses the extended survival analysis method (Shepherd 1999) fitted using the same data and parameters as the stock assessment model of ICES (ICES 2023b).

To run the simulations, the hypothesis made is that the recruitment of sole in the fishing stock at age 2 is dependent on the river flow as highlighted by the delta model. The mean recruitment value R_y is computed as:

$$R_y = \frac{\sum_{i=1}^n \text{predict}_{\Delta}(\text{zone} = z_i, Qa = A_{y-2}) \times c_i}{\sum_{i=1}^n \text{predict}_{\Delta}(\text{zone} = z_i, Qa = 0) \times c_i} \times \bar{R}_{hist} \quad (6)$$

$$A_y = \frac{1}{n_r} \sum_{r=1}^{n_r} Qa_{r,y} \quad (7)$$

with $\bar{R}_{hist}$ the mean historical recruitment computed as the geometric mean of recruitment over the last three years, c_i the relative contribution of nursery z_i in the recruitment of the sole of the Bay of Biscay (Trimoreau et al. 2013), the values of c_i are presented in table 4, A_y the arithmetic mean of the flow anomaly over the four rivers for the year y (eq. 7) and $\text{predict}_{\Delta}(\text{zone}, Qa)$ the CPUE predicted by the delta-lognormal model described in equation 4. All predictions are done for the classes of bathymetry and sediment that maximise the probability of presence, which are the shallowest bathymetry and the muddy sediments. The prediction is run with $Qa = A_{y-2}$ because the delta-lognormal model account for the effect of flow anomaly on age 0 soles but the recruitment in the fishing stock is at age 2. Uncertainty is produced in the recruitment time series by drawing the values in a log-normal distribution centred on R_y and with a standard error equal to the standard error of the 2016-2022 recruitment estimates of the VPA (ICES 2023b).

The stock data is trimmed to remove data after 2016 and the simulation is run over the period 2016 to 2023. This period is chosen as it is only since 2016 that the current management strategy has been used for the stock (ICES 2016; ICES 2023b).

Zone (z_i)	Contribution (c_i)
Vilaine	0.05
Loire	0.06
Bourgneuf	0.13
Pertuis Breton	0.15
Pertuis Antioche	0.44
Gironde	0.17

Table 4: Relative contribution of nursery zones in the Bay of Biscay (Trimoreau et al. 2013).

2.2.3 Testing the hypothesis on recruitment

To be able to run the simulations with the MSE, the assumption that the recruitment of G2 soles in the Bay of Biscay is dependent on the river flow anomaly is made. This assumption is tested by fitting a linear model to assess the effect of river flow anomaly on the number of recruited soles estimated by the virtual population analysis (VPA) model used by ICES to perform the stock assessment which estimates abundance of ages 2 to 8 (ICES 2023b). The VPA is based on commercial indices and one scientific survey index but not all indices provide information on G2. Indeed, only the ORHAGO scientific survey provides reliable data for G2 to the VPA model, the commercial indices do not capture the full G2 abundance and most of the indices provide data starting from age 3 group. Therefore, the age 2 group abundance might not be as well estimated as the older groups and the VPA outputs for G3 are deemed more reliable for this analysis. The last 20 years of estimated recruitment are used in this analysis using data from 1998 to 2018, as for the last three years the recruitment estimates remain uncertain. The beginning of the time series is not used in order to remove the high recruitment values that do not represent the current situation of the stock. The mean of the flow anomaly over the five rivers for the spawning year A_{y-3} for G3 (eq. 7) is used to model the log-transformed abundance of G3 using a linear model. The effect of flow anomaly on the abundance is assessed by testing the difference from 0 of the fitted coefficient using a Ward test.

2.2.4 Management scenarios

Maximum sustainable yield : base scenario

The harvest control rule currently used by ICES is based on the maximum sustainable yield (MSY) approach (ICES 2023b; Anonymous 2019). The MSY is a theoretical point

representing the largest yield that can be taken from the fish stock, over an indefinite period, under constant environmental conditions, while maintaining a maximal growth rate (Maunder 2008). The reference points for the MSY approach for the stock of sole in the Bay of Biscay have been set in 2016 (ICES 2016). The maximum rate of fishing mortality is $F_{MSY} = 0.33$, with a range of $[F_{lower}, F_{upper}] = [0.180, 0.49]$. The minimum biomass threshold is $MSYB_{trigger} = 10600t$, it represents the point below which specific management action needs to be taken to ensure the rebuild of stocks above levels capable of producing MSY in the long term. The limit biomass level under which the stock may not be able to reconstruct itself is $Blim = 7600t$. The management plan uses the F_{MSY} and, in the case of projected biomass under the $MSYB_{trigger}$ threshold, F is decreased proportionally to the ratio $SSB/MSYB_{trigger}$ (equation 8) (ICES 2023b; Anonymous 2019). This management approach is used as the base harvest control rule in the assessment model of the MSE.

$$F = \begin{cases} F_{MSY} & \text{if } SSB > MSYB_{trigger} \\ F_{MSY} \times \frac{SSB_{y+1}}{MSYB_{trigger}} & \text{if } SSB \leq MSYB_{trigger} \end{cases} \quad (8)$$

Environment-based fishing mortality scenarios

To take into account the impact of river flow on the young soles in the estuarine nurseries, several scenarios are defined using river flow to set up an environment-based fishing mortality reference point F_{env} . This approach is based on the ecosystem-based fishing mortality reference point (F_{ECO}) developed by Bentley et al. 2021. The river flow anomaly is used to define an indicator I_y for the year y that reflects the level of flow anomaly relative to a long-time range:

$$I_y = \frac{A_{y-2} - A_{min}}{A_{max} - A_{min}} \quad (9)$$

with A_{y-2} the mean flow anomaly in the Bay of Biscay for the year $y-2$ (eq. 7), A_{min} and A_{max} the minimum and maximum value of A over the year 1982 to 2022. The indicator for year y is computed using the flow anomaly of year $y-2$ because the impact of flow anomaly considered is on the age 0 group of soles that will enter the fishing stock two years later. This indicator I_y determines the value of $F_{env,y}$ in the range $[F_{lower}, F_{upper}]$. If the SSB get lower than the point $MSYB_{trigger}$, $F_{env,y}$ is reduced linearly toward zero

and if SSB reaches the limit B_{lim} , $F_{env,y}$ is automatically set to zero :

$$F_{env,y} = \begin{cases} F_{lower} + I_y \times (F_{upper} - F_{lower}) & \text{if } SSB > MSY B_{trigger} \\ I_y \times F_{MSY} \times \frac{SSB_{y+1}}{MSY B_{trigger}} & \text{if } SSB \leq MSY B_{trigger} \\ 0 & \text{if } SSB \leq B_{lim} \end{cases} \quad (10)$$

A modified version F'_{env} that would be less restrictive for the case $SSB \leq MSY B_{trigger}$ is also tested :

$$F'_{env,y} = \begin{cases} F_{lower} + I_y \times (F_{upper} - F_{lower}) & \text{if } SSB > MSY B_{trigger} \\ (F_{lower} + I_y \times (F_{upper} - F_{lower})) \times \frac{SSB_{y+1}}{MSY B_{trigger}} & \text{if } SSB \leq MSY B_{trigger} \\ 0 & \text{if } SSB \leq B_{lim} \end{cases} \quad (11)$$

These two environment-based fishing mortality reference points are tested as two new scenarios: scenario F_{env} and scenario F'_{env} .

Recruitment computation modifications

An alternative scenario R_{env} affecting the projected recruitment is proposed. In this scenario, the fitted delta model is used to modify the projected recruitment and give advice for the next year according to the river flow anomaly in the Bay of Biscay. Instead of using the geometric mean of past recruitment, the modulated recruitment R_{y+1} and R_{y+2} described by equation 6 are used. For this scenario, the management rule applied is the base MSY rule described before (eq. 8).

2.2.5 Runing the MSE simulations

The MSE is run using packages provided by the FLRproject (Kell et al. 2007) using R version 4.2.2. The VPA model for the operating and assessment model is run using version 2.6.6. of the FLXSA package. The projections are done using a short-term forecast (STF) method performed with version 2.5.11. of the package FLash. Each scenario is run for 200 iterations to account for the variability introduced in the simulation.

3 Results

3.1 Modelling the effect of river flow anomaly on age 0 group CPUE

3.1.1 Binomial presence model

Seven combinations of sampling variables are tested to explain the variability in the presence of G0 soles and compared with the null model (table 5). Among the three variables used individually, bathymetry is the variable which is responsible for the greater reduction in the AIC value compared to the null model. The combination of variables resulting in the lowest AIC value is the full model.

Terms	AIC	ΔAIC
null	3770	0
sediment	3687	-83
bathymetry	3583	-187
zone	3727	-43
bathymetry + sediment	3476	-294
zone + sediment	3642	-128
zone + bathymetry	3542	-228
zone + bathymetry + sediment	3420	-350

Table 5: Explanatory variables combinations comparison for the binomial presence model. For each fitted model the AIC is displayed as well as the ΔAIC with the null model presented in the first line.

The addition of the river flow anomaly results in a statistically significant (p-value = $3.0e^{-10}$) increase in the deviance explained by the model compared to the model without the flow anomaly (table 6). Adding the interaction $Qa \times zone$ does also improve the model, but with lower significance (p-value = $3.6e^{-2}$) and a very small decrease of 2 points in AIC. None of the fitted coefficients for the interaction is significant, hence the interaction term is not be kept in the model. The model with the zone, the bathymetry category, the sediment type and the flow anomaly is selected to model the probability of presence. We can conclude a significant effect of the flow anomaly on the probability of presence of G0 soles.

The fitted coefficients for the binomial model depict an increased probability of presence the shallower the water gets (figure 4). The difference between the reference class of

Terms	Resid. Dev	Df	Deviance	Pr(>Chi)	AIC
zone + bathymetry + sediment	3394				3420
zone + bathymetry + sediment + Qa	3355	1	40	$3.0e^{-10}$	3383
zone + bathymetry + sediment + Qa + zone $\times$ Qa	3343	5	12	$3.6e^{-2}$	3381

Table 6: Analysis of variance for the binomial presence model. The addition of the flow anomaly and then the interaction between the zone and the flow anomaly are tested.

bathymetry deeper than $-20m$ and all the other classes is statistically significant. The coefficients associated with sediment type display an increased probability of presence of the young soles in muddy and sandy sediments compared with coarse sediments. No significant difference is detected between the mixed sediment class and the coarse sediment class. Only the Pertuis Breton and Vilaine Bay zones are associated with a significant difference in presence probability compared to the reference zone Bourgneuf Bay. In both Pertuis Breton and Vilaine Bay zones, the presence probability is lower than in Bourgneuf Bay. A statistically significant ($p\text{-value} < 0.001$) positive effect of the river flow anomaly on the probability of presence is detected and the associated coefficient is equal to 0.27 on the logit scale. This model explains 11% of the presence data variability.

3.1.2 Log-normal positive density model

For the log-normal model of the positive sole density, the conclusions are similar. The most substantial decrease in AIC value is produced by the bathymetry (table 7). The model that includes all the explanatory variables is selected.

Terms	AIC	ΔAIC
null	5531	0
sediment	5523	-8
bathymetry	5427	-104
zone	5500	-31
bathymetry + sediment	5416	-115
zone + sediment	5485	-46
zone + bathymetry	5402	-129
zone + bathymetry + sediment	5382	-150

Table 7: Explanatory variables combinations comparison for the log-normal CPUE model. For each fitted model the AIC is displayed as well as the ΔAIC with the null model presented in the first line.

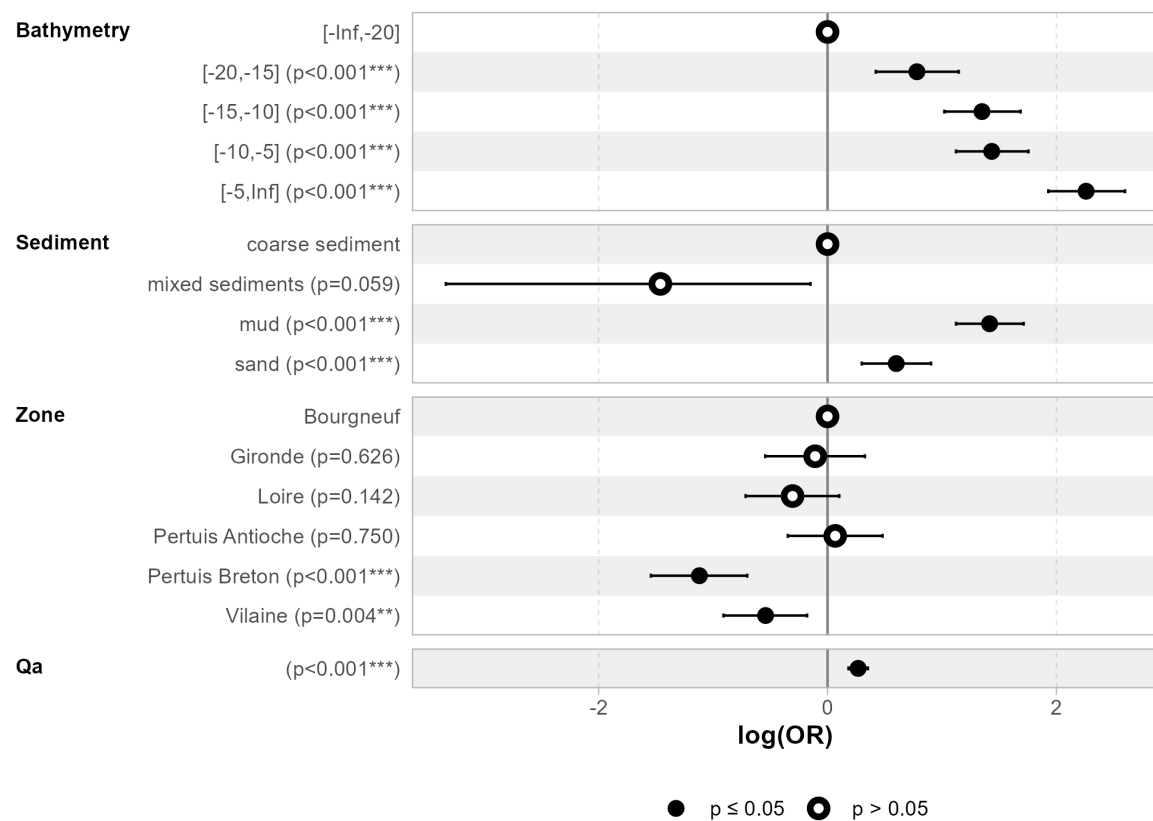


Figure 4: Values of the fitted coefficients of the Binomial presence model. The values are expressed on the logit scale. The first class of each categorical variable is the reference class.

The addition of the flow anomaly to the model with habitat variables improves the model significantly ($p\text{-value} = 1.6e^{-9}$) (table 8). Adding the interaction does not decrease the residual deviance significantly ($p\text{-value} = 0.54$) and the interaction will not be kept in the model. The model including the zone, the bathymetry class, the sediment type, and the flow anomaly is selected to model the positive sole density. And we can conclude on a significant effect of the flow anomaly on the density of G0 soles, with no modification of the effect depending on the sampled zone.

Terms	Resid. Dev	Df	Deviance	F	Pr(>F)	AIC
zone + bathymetry + sediment	3250					5382
zone + bathymetry + sediment + Qa	3169	1	80	37	$1.6e^{-9}$	5347
zone + bathymetry + sediment + zone $\times$ Qa	3161	5	9	1	$5.4e^{-1}$	5353

Table 8: Analysis of variance for the log-normal CPUE model. The addition of the flow anomaly and then the interaction between the zone and the flow anomaly are tested.

Only the bathymetry class $-5m$ and shallower waters is statistically different from the

reference class deeper than $-20m$ (figure 5). As for the presence probability, the density is significantly higher in muddy and sandy sediments compared to coarse sediments. The difference in density between coarse sediment and mixed sediment classes is significant ($p - value = 0.048$). However, the p-value is very close to the threshold value of 0.05 and the confidence interval is very large so no conclusion can be drawn from this result. The mean density in all zones is significantly lower than in Bourgneuf Bay. The shallower waters are associated with higher densities of G0 soles. The effect of river flow anomaly is positive and statistically significant ($p - value < 0.001$) and with a coefficient of 1.25 in the CPUE scale. This model explains 13.3% of the deviance in the strictly positive CPUE values of the data.

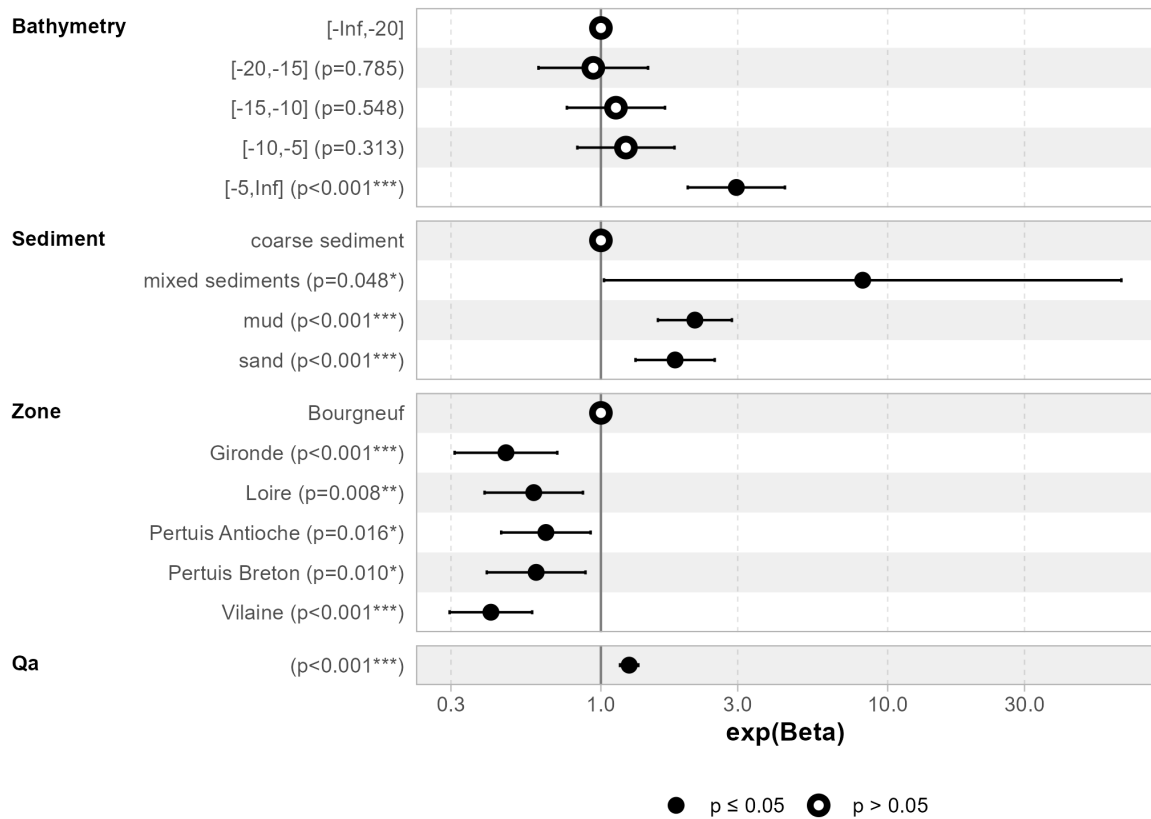


Figure 5: Values of the fitted coefficients of the log-normal density model. The values are expressed on the CPUE scale. The first class of each categorical variable is the reference class.

3.2 Testing the relationship between the river flow anomaly and the G3 abundance estimates

The fitted coefficient of regression between the log-transformed abundance of G3 estimated by the VPA and the mean river flow anomaly is equal to 0.116 on the log abundance scale. The coefficient is significantly different from zero with risk 5% ($p\text{-value} = 0.0478$). This model explains 25.1% of the deviance in the G3 estimations.

3.3 Management strategy scenarios evaluation

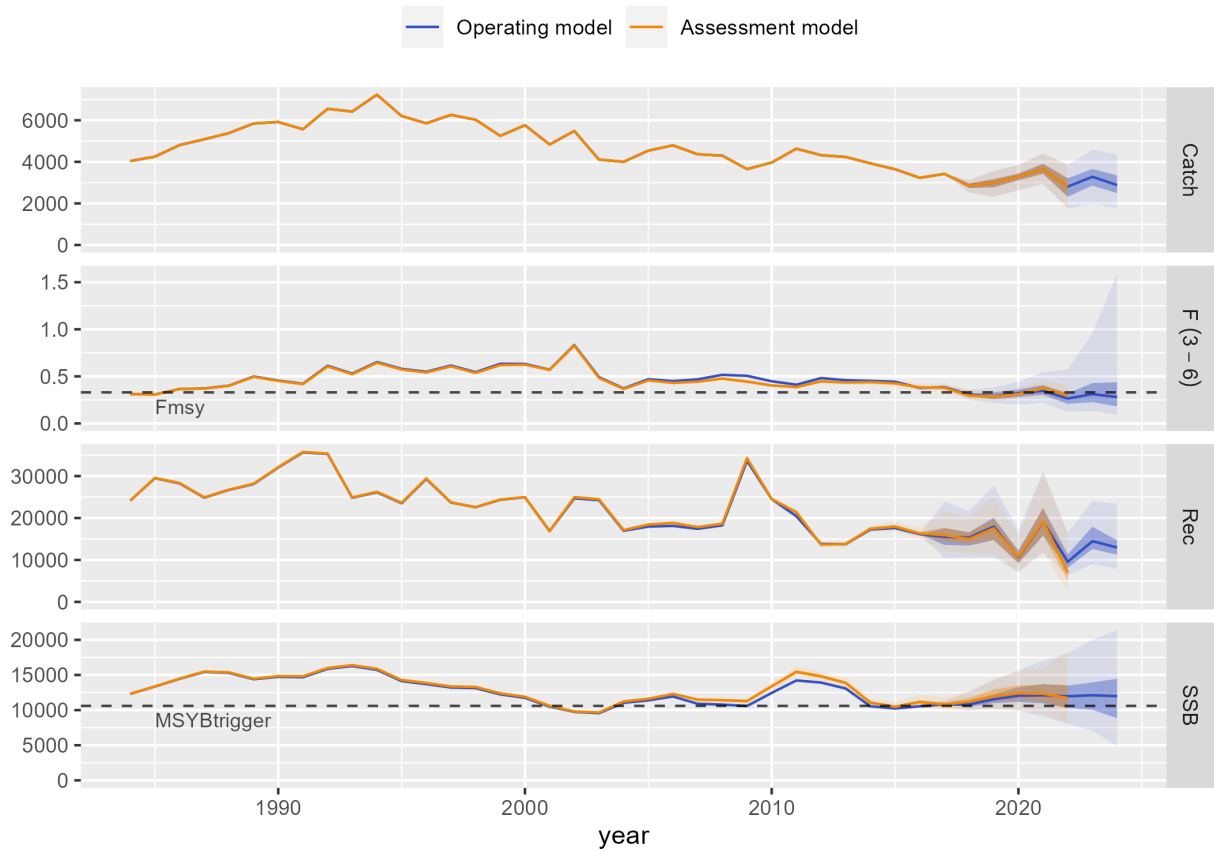


Figure 6: Comparison of the operating (purple) and assessment (orange) models for the base scenario over the whole time series. The lines represent the median values of each metric over the 200 iterations and the coloured ribbons are the quantiles 25-75% and 5-95%. The metrics are the amount of catch per year in tonnes, the average fishing mortality over 3 to 6 age groups (F), the recruitment at age 2 (Rec) and the spawning stock biomass (SSB).

The results of the MSE runs for the base scenario are displayed in figure 6. As expected for the base scenario that uses the MSY rule, the median F along the simulated years stays close to the target F_{MSY} (fig. 6 and 7, panel F). The variability included in the

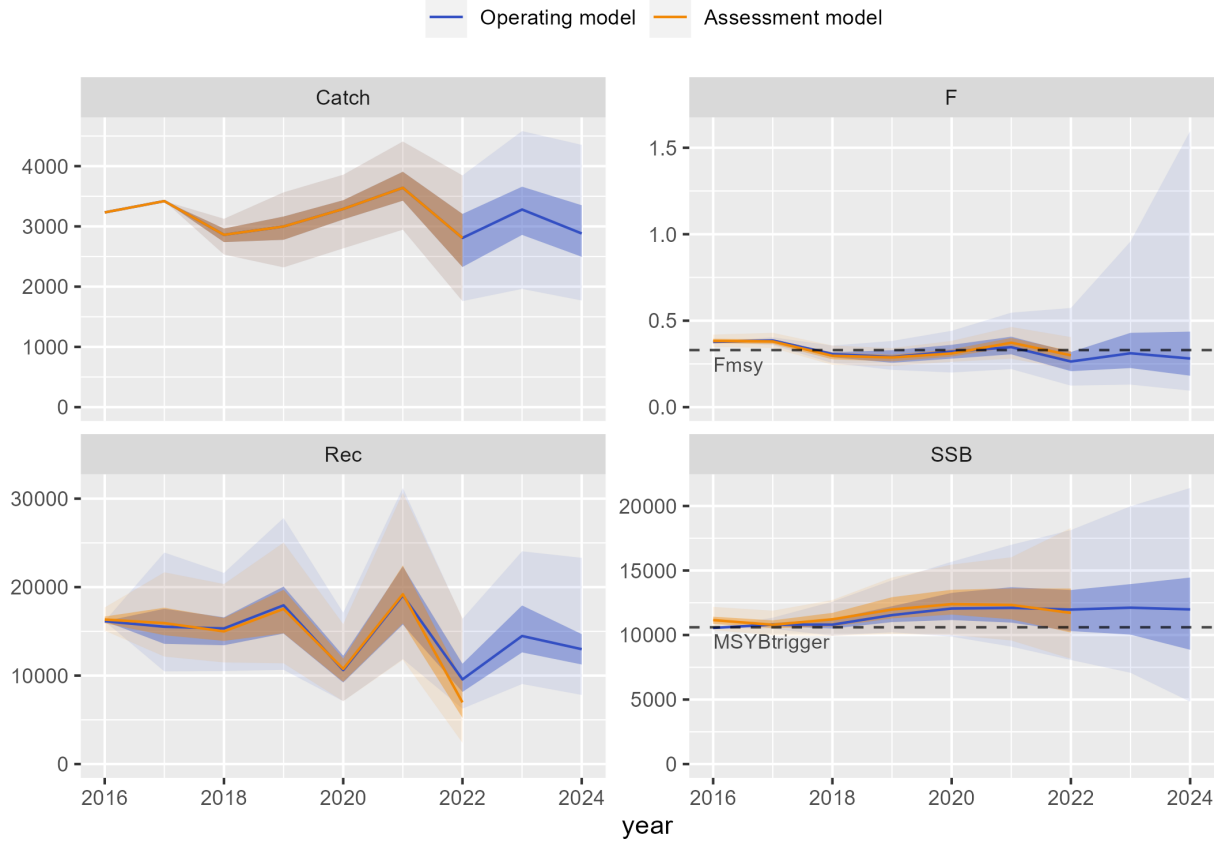


Figure 7: Comparison of the operating (purple) and assessment (orange) models for the base scenario over the years 2016 to 2024. The lines represent the median values of each metric over the 200 iterations and the coloured ribbons are the quantiles 25-75% and 5-95%. The metrics are the amount of catch per year in tonnes, the average fishing mortality over 3 to 6 age groups (F), the recruitment at age 2 (Rec) and the spawning stock biomass (SSB).

MSE is responsible for the increase of dispersion of F and SSB along the simulated years. The modulation of the simulated recruitment by the flow anomaly in the operating model can clearly be seen in panel Rec of figure 6 and is well estimated by the assessment model (fig. 7). The median SSB values rise above the $MSY B_{trigger}$ and stabilise itself however it does not reach historical levels that were generally higher before the years 2000's (fig. 6, panel SSB). A difference between the operating model simulations and the estimated SSB by the assessment model can be noted, with an overestimation of the SSB between 2018 and 2021 and an underestimation for the last data year 2022 (fig. 7, panel SSB). The catch values in the operating and assessment model are exactly the same (fig. 6, panel Catch) as designed in the MSE implementation, the TAC set with the assessment model is exactly reached in the operating model.

The scenarios F_{env} and F'_{env} share similar median TAC values across the simulated

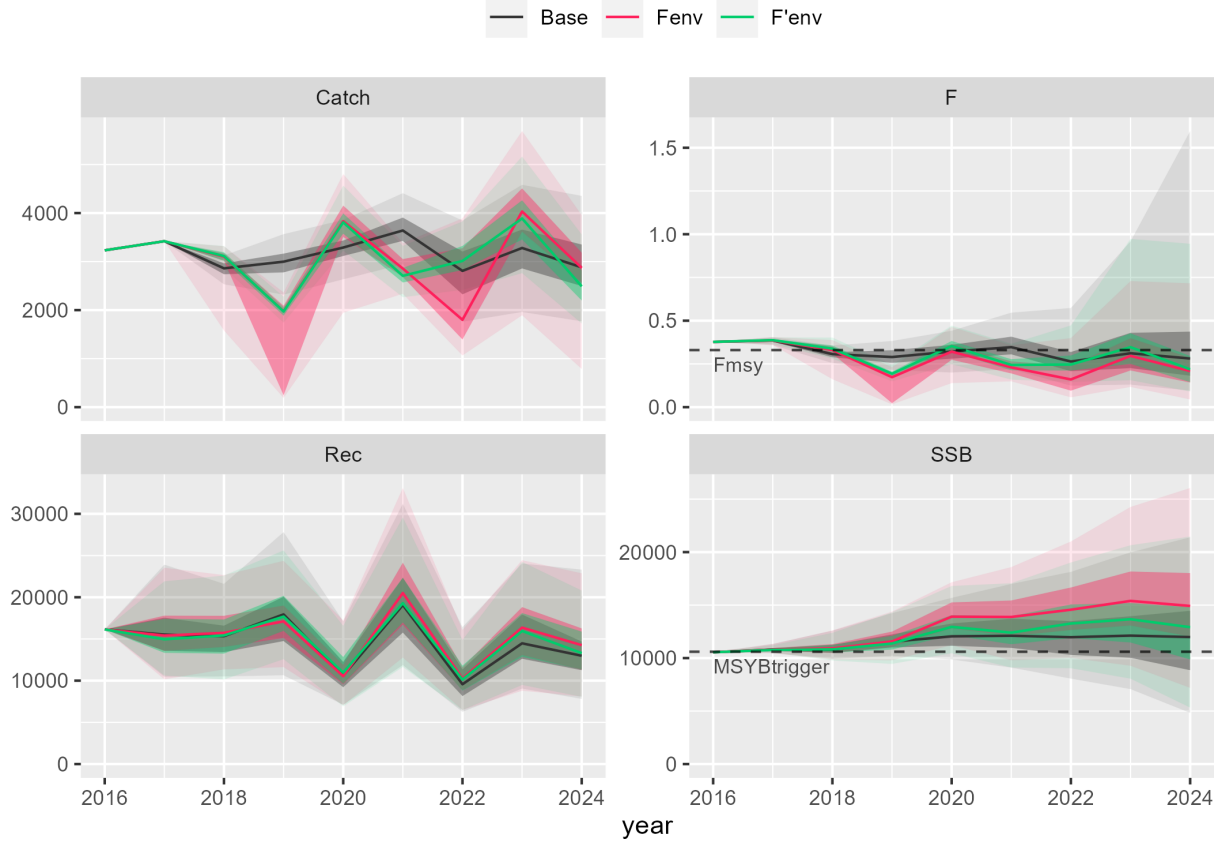


Figure 8: Comparison of the operating model between the base scenario (grey), F_{env} (red) and F'_{env} (green) over the years 2016 to 2024. The lines represent the median values of each metric over the 200 iterations and the coloured ribbons are the quantiles 25-75% and 5-95%. The metrics are the amount of catch per year in tonnes, the average fishing mortality over 3 to 6 age groups (F), the recruitment at age 2 (Rec) and the spawning stock biomass (SSB).

years but the quantiles intervals are higher for the F_{env} scenario (fig. 8, panel Catch). This is caused by the case $SSB \leq MSYB_{trigger}$ that differs between the scenarios (equations 10 and 11) with scenario F_{env} producing very low F values in this case compared to F'_{env} scenario. For these scenarios, the difference in the evolution of SSB in the operating model is clearly visible (fig. 8, panel SSB). Both F_{env} and F'_{env} result in a greater increase of SSB compared to the base scenario. The F_{env} scenario produces a bigger increase in median SSB for the year 2020 when some TAC values were greatly decreased by the decline of the projected SSB under the threshold $SSB < MSYB_{trigger}$.

The use of the R_{env} scenario induces a clear change in the TAC advised, especially for the years 2020, 2021 and 2022 (figure 9, panel Catch). These years are characterised by clear variations in the simulated recruitment, driven by the flow anomaly of year $y - 2$. In 2020 and 2022 the simulated recruitment is lower than average (fig. 9, panel

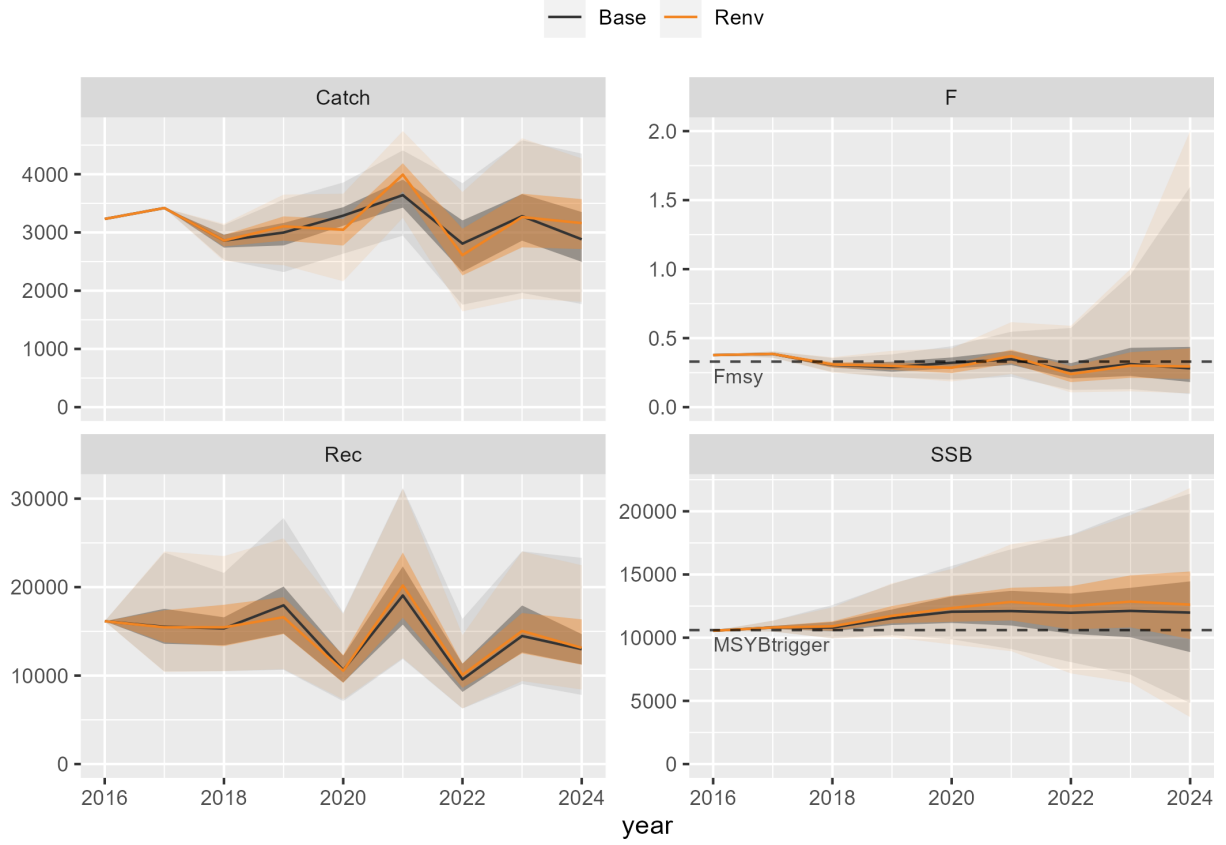


Figure 9: Comparison of the operating model between the base scenario (grey) and R_{env} (orange) over the years 2016 to 2024. The lines represent the median values of each metric over the 200 iterations and the coloured ribbons are the quantiles 25-75% and 5-95%. The metrics are the amount of catch per year in tonnes, the average fishing mortality over 3 to 6 age groups (F), the recruitment at age 2 (Rec) and the spawning stock biomass (SSB).

Rec) and the response of the environmentally informed recruitment scenario can be seen immediately in the low TAC values set for these years. The response is also visible for higher recruitment, such as for the year 2021, which induces higher TAC for this year. However, these differences in TAC between the base scenario and the R_{env} scenario do not induce a clear difference in the simulated SSB of the operating model (fig. 9, panel SSB).

The modification of the harvest control rule according to the fishing mortality scenarios F_{env} and F'_{env} produces more important changes in the simulation results compared to the R_{env} scenario (figure 10). The F_{env} and F'_{env} scenarios generate more variable advice on catch compared to the R_{env} scenario which is closer to the base scenario (fig. 10, panel Catch). The more drastic changes in the TAC simulated for F_{env} and F'_{env} create a greater increase in the operating model SSB compared to R_{env} (fig. 10, panel SSB).

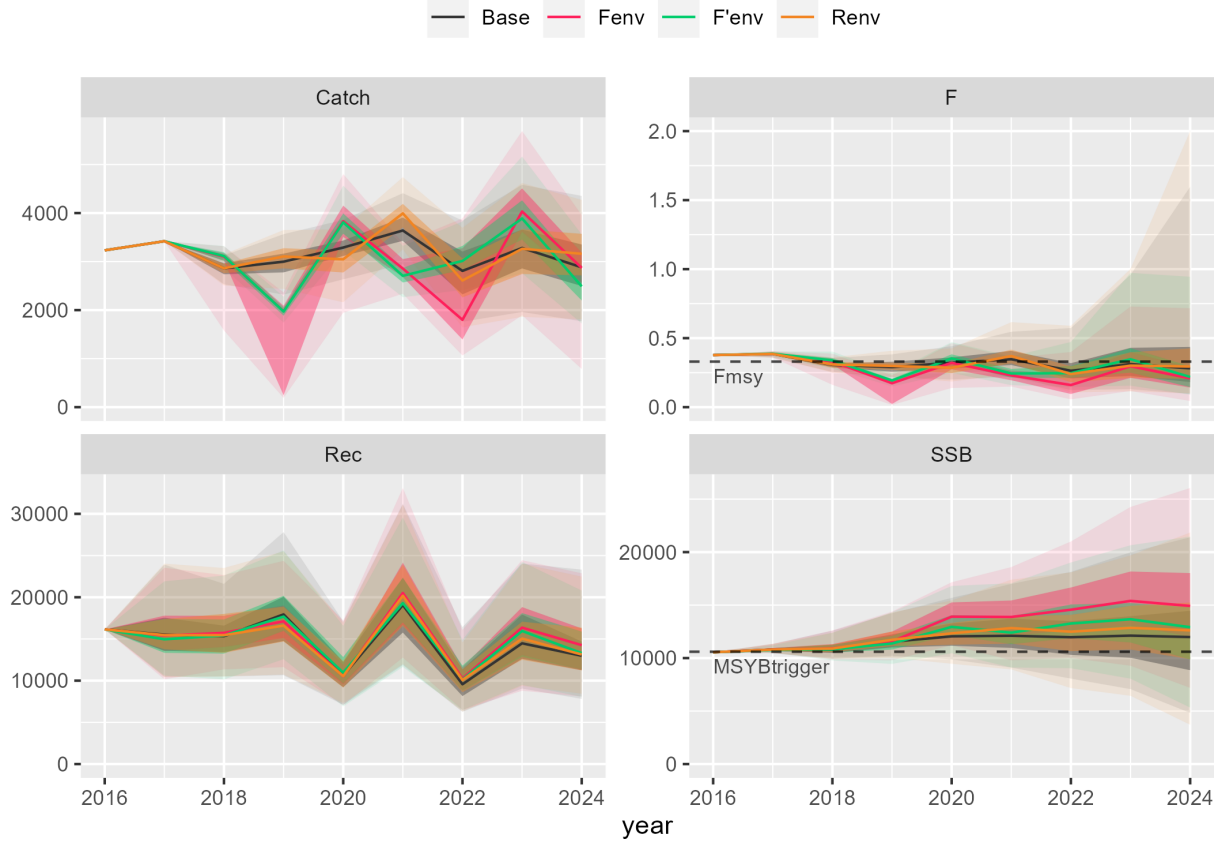


Figure 10: Operating model comparison between the base scenario and the three alternative scenarios defined over the years 2016 to 2024. The line represents the median values of each metric over the 200 iterations and the coloured ribbons are the quantiles 25-75% and 5-95%. The metrics are the amount of catch per year in tonnes, the average fishing mortality over all age groups (F), the recruitment at age 2 (Rec) and the spawning stock biomass (SSB).

4 Discussion

4.1 Relationship between the river flow anomaly and the abundance of G0 soles

The first approach in this study consisted of using the NURSE survey data to highlight the relationship between recruits of sole and river flows in the Bay of Biscay by using a model that also includes habitat description variables. The model captures the known preferences for specific habitat types already described for the sole. Indeed, our approach highlights the importance of shallow water areas in providing nursery grounds for G0 sole as described in Le Pape, Chauvet, et al. 2003 and Trimoreau et al. 2013. Soles are also known to prefer fine sediments, especially muddy sediments (Le Pape, Chauvet, et al. 2003; Gibson and Robb 2000; Trimoreau et al. 2013), which is also captured by the

delta-lognormal model.

The approach also succeeded in highlighting the relationship between the river flow anomaly and the abundance of age 0 soles. The results obtained with the delta-lognormal model do conclude on a positive effect of river flow anomaly. This result supports what has already been observed for the relationship between the extent of the river plume in Vilaine Bay and the extent of nursery grounds (Le Pape et al. 2003), the relationship between age 1 soles and the flow anomaly in the Vilaine Bay and between age 2 soles and the flow anomaly in the Bay of Biscay (Le Pape, Désaunay, and Guérault 2003).

The sediment types are grouped into three categories in order to minimise the complexity of the model by reducing the number of coefficients to fit while keeping meaningful categories in regard of the European Nature Information System (EUNIS) classification. Nonetheless, the mixed sediments category results in a very small category with only 21 hauls. The small amount of data in this category explains the very large error on the associated fitted coefficient. This category should be modified, grouped with another one or dropped and the associated coefficient cannot be interpreted as it is. However, this category is not used in the rest of the study so it does not impact the rest of the results.

The choice of using the anomaly of the mean river flow over the first month of the year can be discussed. Firstly, the choice of the five first months of the year is an empirical choice motivated by the observation made on the flow data and on the knowledge of the life-cycle of the sole. This choice can have an impact on the associated fitted coefficient and a sensibility analysis could be made to assess the magnitude of variation in the results obtained when modifying the number of months used to compute the flow anomaly. Furthermore, other metrics should be investigated to better understand the relationship between the abundance of young soles and the freshwater inputs in nurseries. For example, metrics related to river flood events could be interesting to analyse as floods play an important role in nutrient export from land to the ocean. River flood events can be summarised by several metrics, the number of flooding per season but also the frequency of the flood, the amplitude or the period of the year. This type of analysis could lead to more insight into how the freshwater input of the rivers impacts the coastal habitat.

The delta-lognormal only captures 11% of the variability of presence data and 13.3% of the variability of strictly positive CPUE data. The majority of variability is not explained by the variables included in the study. Other variables should be explored to improve the

knowledge of the phenomena that impact the abundance of soles during their first year and permit a better description of the G0 abundance. Other parameters that could impact the G0 soles in the nurseries are the water quality or the availability and quality of preys in the environment for example. The variability of sole abundance in the nursery may also be significantly impacted by potential changes in the hydroclimatic conditions, impacting the earlier egg and larval life stage (Rijnsdorp et al. 1992; Savina et al. 2016).

4.2 Alternative management strategies and MSE simulations

To run simulations with the MSE loop, a hypothesis on simulated recruitment values was necessary as no stock-recruitment relationship is defined for this stock. The hypothesis made is that recruitment is influenced by the river flow anomaly of the spawning year according to the fitted delta-lognormal model. The projection requires an input of data for the recruitment at age 2, however, the quality of the estimated abundance of G2 soles by the VPA was not considered reliable enough so the G3 abundance was used for this analysis. The result of the analysis did conclude on a positive effect of river flow on the abundance of G3 soles. Based on this results, the recruitment for the operating model was simulated using the relationship with river flow anomaly described previously. However, this hypothesis remains uncertain and the results of the MSE simulations must be seen only as a theoretical view of how different management strategies could impact the stock. This analysis highlights the necessity of further research on the factors impacting the soles during their first years in order to better understand the decrease in recruits that has been happening for a few years.

The results of the MSE simulations represent the projection of the stock under four different scenarios: the base scenario corresponding to the current management strategy for this stock and three alternative scenarios, one that modifies the projected recruitment and two that modifies the harvest control rule.

The approach of environment-modulated recruitment value R_{env} does simulate changes in advice compared to the base scenario but it is not sufficient to induce a clear difference in the spawning stock biomass evolution. This scenario is more cautious than the base scenario as it anticipates lower recruitment by reducing the TAC.

The two harvest control rules proposed here are based on the same environmental indicator but the simulations run show that they can lead to different results. Depending

on the objective set for the stock evolution, different strategies should be considered (Bentley et al. 2021; Punt et al. 2016). The scenario F_{env} has a very restrictive rule when the SSB gets lower than the MSY biomass trigger point and is very protective against the decline of the SSB. However, the TAC values resulting from this scenario are highly variable, with years where the fishery should almost be closed which would not be desirable for the actors of the fishing sector (Patterson et al. 2007). The F'_{env} scenario is a more moderate approach. The rule for F'_{env} in the case $SSB \leq MSY B_{trigger}$ is less restrictive compared to the one of F_{env} . The value of F does not change drastically when the SSB gets lower than the MSY biomass trigger point so it results in less extreme low TACs compared to F_{env} . With this more moderate approach F'_{env} , the SSB does not increase as much as with the other scenario F_{env} but it still increases above the SSB level of the base scenario and may be easier to accept for the fishing sector to the F_{env} scenario and its extremely low catch advice.

The MSE results are a theoretical approach to how management strategies could be modified and their impact on the stock. It cannot provide a realistic view of the evolution of the stock as we do not have enough knowledge of the driver processes affecting the recruitment to be able to model it realistically. Further analysis could be made, taking other environmental indicators, such as water quality indicators, to study their impact on recruitment and eventually developing other management scenarios.

In the case that one of the proposed scenarios based on river flow anomaly would be used for stock management, new river flow data would be necessary each year. The river flow variations impact the soles during their first year but the recruitment in the fishing stock is at age 2 so the data are needed with a two years lag. The availability of this data should not be an issue as the flow data are quickly made available on the HydroPortail. However, the relationship between recruitment and the river flow might change or disappear over time. The relationship should be tested every year and the management plan should go back to using the base MSY strategy in the case of a loss of this relationship.

5 Conclusion

This study highlights the impact of the river flow variation in nursery grounds on the abundance of age 0 group of soles as well as the already described preference of young soles for certain habitat types. Based on this first analysis, the impact of using river flow variation in the management of the sole stock in the Bay of Biscay was assessed. The recruitment of the stock was simulated using the relationship described by the model accounting for river flow variation and habitat preferences. Alternative management strategies were defined to take the results of the first analysis into account. In order to test and compare these alternative scenarios with the current management strategy, a MSE framework was developed. The three alternative strategies were all based on the same metric of river flow anomaly. In one scenario, the projected recruitment was modified to vary according to the river flow anomaly of the spawning year. For the two other scenarios, the harvest control rule was changed to take into account the variations of river flow anomaly of the spawning year to set the target fishing mortality. The results of the MSE simulations for the different alternative scenarios showed modifications of advised TAC and an increase in SSB compared to the base scenario. This study highlights the impact of the harvest control rule chosen to manage stock and the importance of defining the objective to reach for the stock in order to be able to adapt the harvest control rule used.

The results of this study will be used for the benchmark of the assessment model planned by ICES for February 2024. The current assessment model will be changed to a stochastic model that will probably model the stock starting at age 1. In this case, supplementary data will be needed to evaluate the abundance of age 1 group. The delta-lognormal model fitted will be used to produce an abundance index from the NURSE survey.

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A Hydrological stations

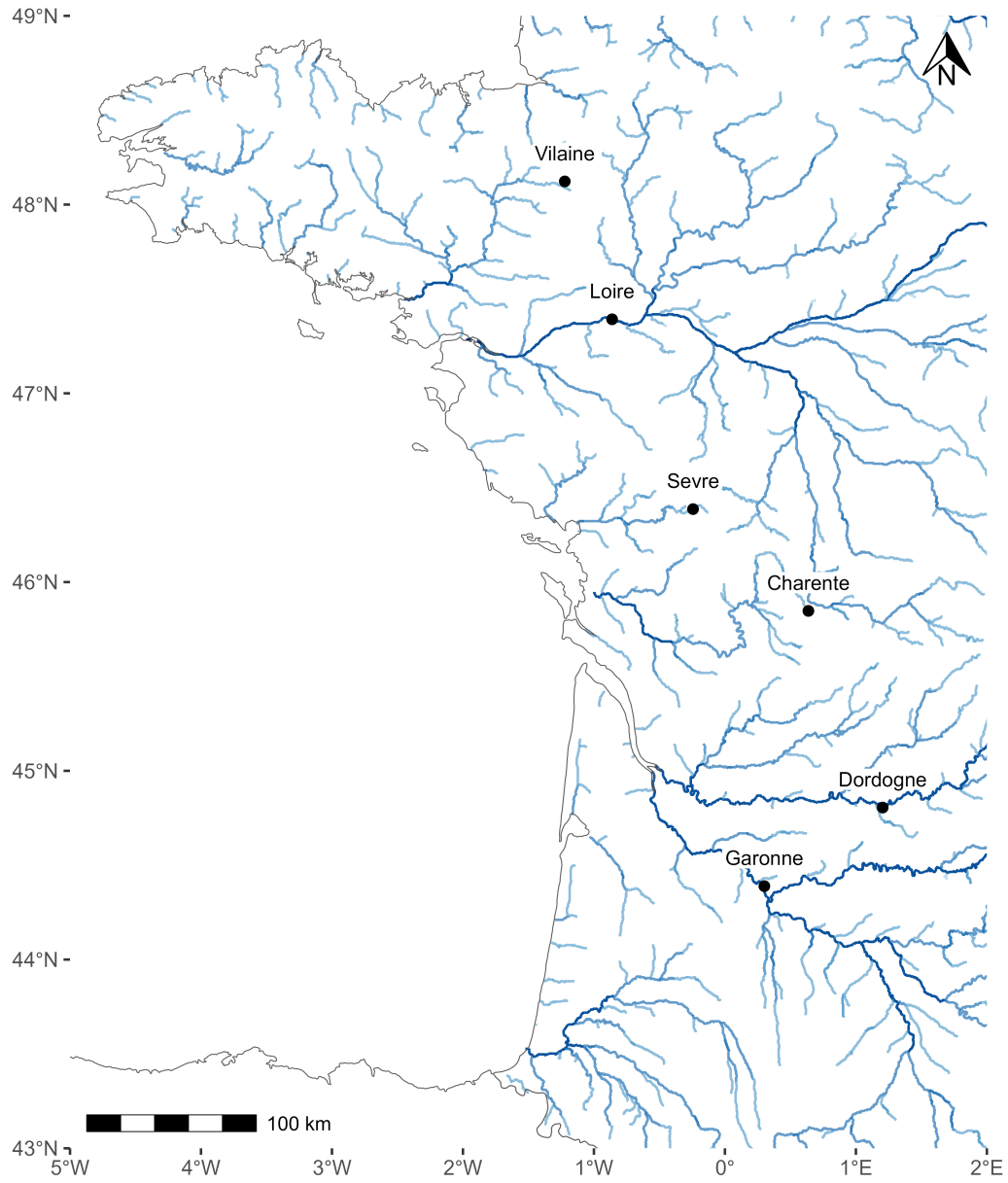


Figure 11: Map of the main French rivers flowing in the Bay of Biscay. The position of the hydrological stations for which the monthly flow data is retrieved is shown and labelled with the name of the corresponding river.

River	Code
Vilaine	J7010610
Loire	M5300010
Sevre	N4010610
Garonne	O900001002
Dordogne	P2380010
Charente	R0020010

Table 9: Banque Hydro hydrometric entity identifier code of the stations that were used to obtain the monthly flow data for each river.

B River flow data

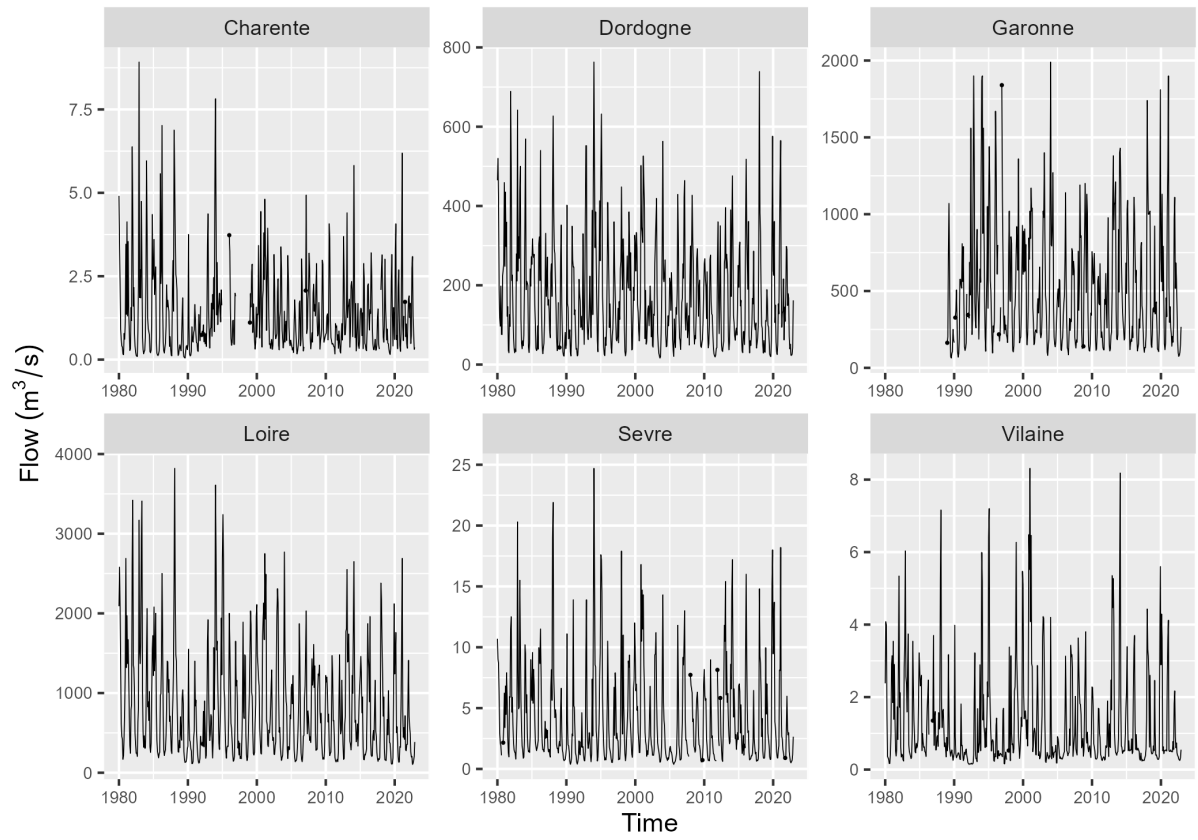


Figure 12: Monthly mean river flow data for each river over the period 1980-2022.